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Physics of the Earth and Planetary Interiors 108 (1998) 291–303

PHYSICS OF THE EARTH AND PLANETARY INTERIORS

Paleomagnetism of Pleistocene volcanic rocks from Pantelleria Island (Sicily Channel), Italy

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Received 6 January 1998; revised 19 May 1998; accepted 29 May 1998

Abstract

A paleomagnetic investigation has been carried out on Pleistocene volcanic rocks from Pantelleria, in the Sicily Channel. This island is characterised by a bimodal volcanism of peralkaline rhyolite to trachyte and basalt composition. The radiometric data indicate that volcanic activity started around 324 ka BP. Samples have been collected from 30 sites in 16 volcanic units spanning activity during the last 150 ka. Magnetic properties vary systematically with lithotype and indicate high-Ti titanomagnetite to magnetite as the main carriers of magnetisation. Stable characteristic remanent magnetisation (ChRM) directions isolated by alternating fields (Af) demagnetisation show normal polarity consistent with emplacement during the Brunhes chron and vary within the paleosecular variation (PSV) range. However, the inclinations are low as compared with the geocentric axial dipole (GAD) inclination at Pantelleria, especially for volcanic units younger than 50 ka. The mean ChRM direction computed from 27 sites is $D = 358.8°$, $I = 46.9°$ ($k = 30$, $\alpha_{95} = 5.2°$), and the inclination anomaly is $\Delta I = -8.1°$. Both tectonic movements and the presence of large magnetic anomalies around the island of Pantelleria fail to explain this low inclination, which may therefore be related to a significant long-lived, non-dipolar field contribution in the area over the past 150 kyr. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: Paleomagnetism; Pleistocene volcanic rocks; Pantelleria Island

1. Introduction

Paleomagnetic directions recorded in volcanic rocks have often been used to investigate paleosecular variation (PSV) of the geomagnetic field, and lava flow databases have been compiled to model the time-averaged geomagnetic field (Quidelleur et al., 1994; Johnson and Constable, 1995, 1996; Kelly and Gubbins, 1997). When we have paleomagnetic records such as lavas able to record instantaneous values of the geomagnetic field direction, if several paleomagnetic determinations are available from a single area, secular changes in the geomagnetic field can be averaged out to obtain a mean value for comparison with the local geocentric axial dipole (GAD) direction (Merrill and McElhinny, 1983). The temporal sampling must span at least 10^5 yr (Johnson and Constable, 1995) and the independence of the flows must be established to avoid a bias in the local estimate of the time-averaged field. Significant non-dipole field features have been detected in the time-averaged field, and long-period variations in the

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PII: S0031-9201(98)00108-3

geomagnetic field have been observed in paleomagnetic records from marine sediments and lava flows: in marine sediment cores from the western equatorial Pacific, the magnetic inclination records have shown a long-term secular variation of 40–50 kyr in periodicity which may be the consequence of orbital forces (Yamazaki and Ioka, 1994); in lava flows, a study of a core from Hawaii for the last 400 kyr displayed systematically low inclinations throughout the sequence (Holt et al., 1996). Additional paleomagnetic data from distributed sites may improve the knowledge of these long-period variations.

Paleomagnetic investigation of the island of Pantelleria, a Pleistocenic composite volcanic area, for

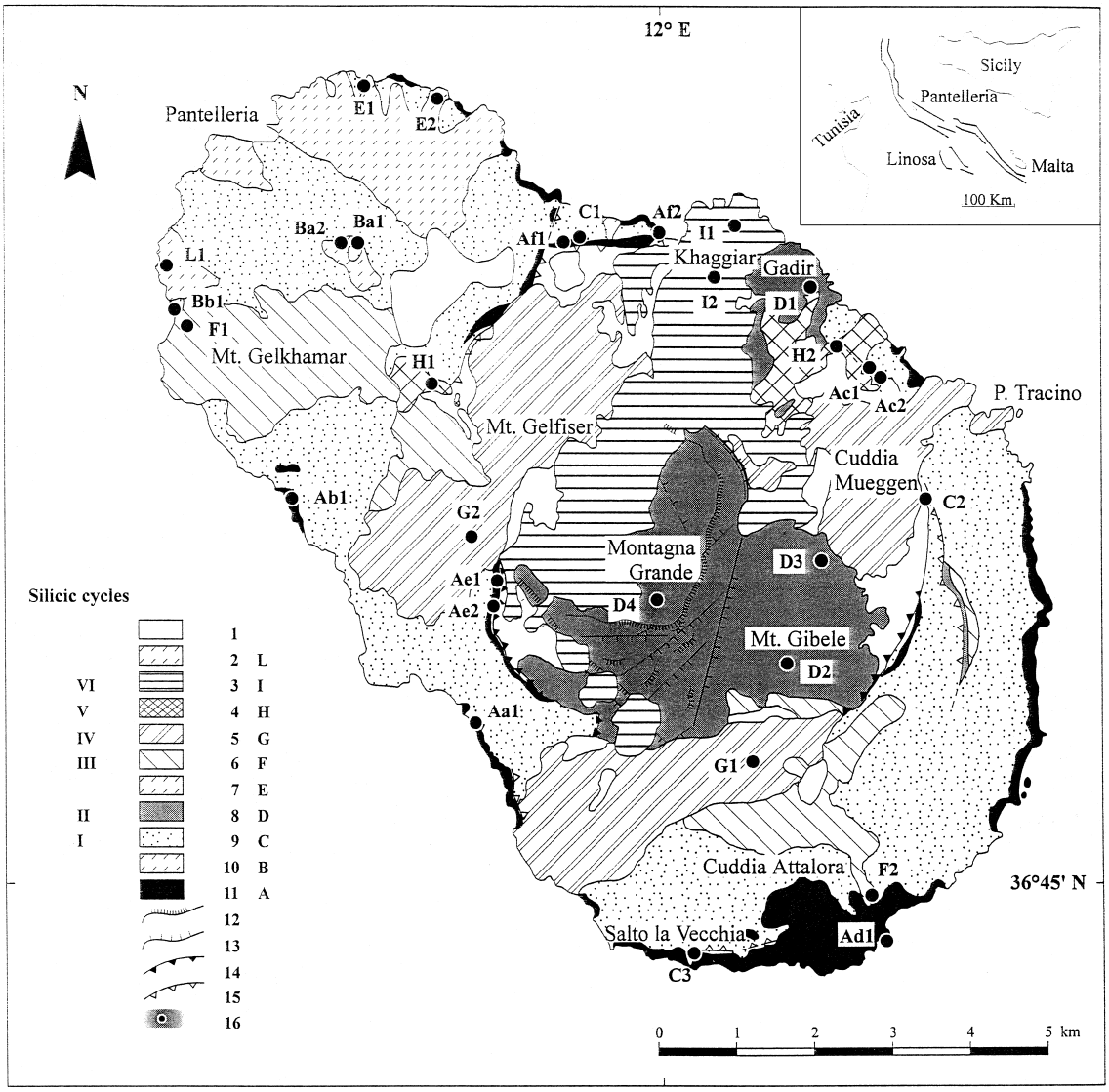


Fig. 1. Geological sketch map of the island of Pantelleria (simplified after Orsi et al., 1991b). Legend: (1) alluvium; (2) Mursia basalt (younger than 10 ka); (3) VI silicic cycle (10–8 ka); (4) V silicic cycle (14–12 ka); (5) IV silicic cycle (20–15 ka); (6) III silicic cycle (ca. 22 ka); (7) Punta San Leonardo basalt (27 ka); (8) II silicic cycle (35–29 ka); (9) I silicic cycle—Green Tuff (50 ka); (10) pre-Green Tuff basalt (120 and 80 ka); (11) pre-Green Tuff silicic volcanites (324–50 ka); (12) top of the volcano-tectonic escarpment; (13) volcano-tectonic faults; (14) Monastero or Cinque Denti caldera rim; (15) La Vecchia caldera rim; (16) sampling sites.

which many radiometric determinations and stratigraphical studies are available, has provided 16 new time-independent paleomagnetic directions for the Mediterranean zone.

2. Geological setting and sampling

Pantelleria is located in the Mediterranean, along the continental rift of the Sicily Channel (Fig. 1). It has been the site of intense volcano-tectonic activity and is entirely composed of Pleistocenic volcanic rocks younger than 324 ± 10 ka (Mahood and Hildreth, 1986). The eruptive products are characterised by peralkaline composition and mainly vary from pantellerites to less peralkaline rhyolites or trachytes, with minor basalts (Civetta et al., 1984).

The island's main volcano-tectonic features are two calderas of different age that form its central part, and a resurgent block inside the younger caldera (Orsi et al., 1991a). 'La Vecchia' Caldera is the older one and formed around 114 ka BP. The great eruption of the Green Tuff eventually covered the whole island at 49.0 ± 2.0 ka (Cornette et al., 1983) and was followed by the collapse of the younger caldera, called either 'Monastero' (Cornette et al., 1983) or 'Cinque Denti' (Mahood and Hildreth, 1983). Gravity and geomagnetic surveys (Gantar et al., 1961; Berrino and Capuano, 1995) suggest that there is a dome-shaped intrusion under the island, and a geothermal exploration drill has found a comendite body at a depth of about 400 m (Pignatelli, 1994).

The products of the Green Tuff eruption are a clear stratigraphic marker and have been used to divide the eruptive history of Pantelleria into two main periods (Cornette et al., 1983). The pre-Green Tuff period was characterised by silicic eruptions. Their products consist of lava flows, lava domes and pyroclastic deposits, which mainly crop out along the sea cliffs and the caldera walls. Two minor basaltic episodes occurred at 118.5 ± 9.0 and 83 ± 5.2 ka in the northern part of the island (Civetta et al., 1984). The second period has been divided into six major eruptive cycles (Civetta et al., 1988) on the grounds of stratigraphical and geochronological data, occurrence of paleosoils and location of the source vents. It began with the eruption of the Green Tuff, which includes ignimbrites, fall and surge deposits whose compositions range from pantellerite to trachyte (Cornette et al., 1983). Most of the activities of the subsequent five cycles were confined within the younger caldera. K–Ar determinations show different ages for each cycle (Orsi et al., 1991b). The

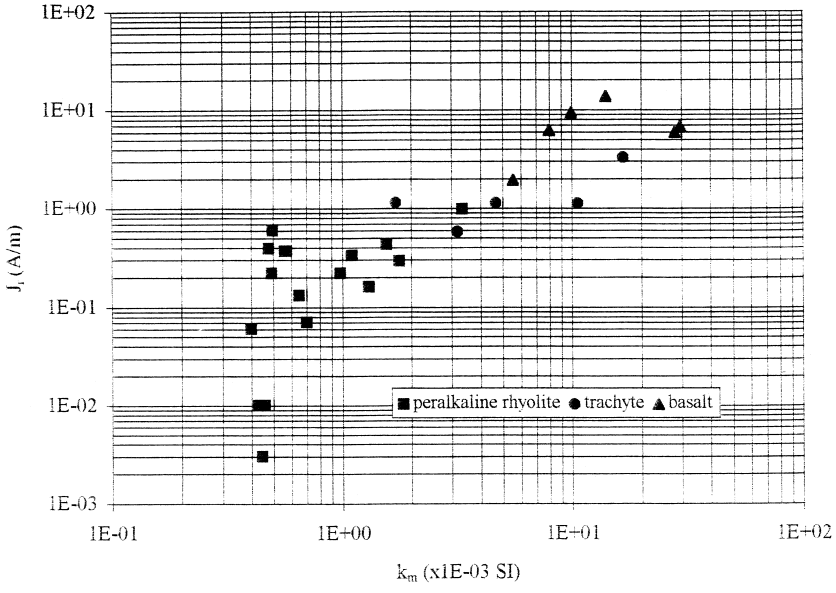


Fig. 2. Bilogarithmic diagram of NRM intensity vs. magnetic susceptibility.

second cycle lasted from 35 to 29 ka, the third occurred at about 22 ka and the forth, fifth and sixth lasted from 20 to 15 ka, 14 to 12 ka and 10 to 8 ka, respectively. They all consist of pyroclastic-fall and pumice-fall deposits, lava domes and lava flows. Besides the silicic activity, two basaltic episodes occurred in the north–western part of Pantelleria outside the calderas (Civetta et al., 1988). The first (Punta San Leonardo Basalt) occurred around 27 ka BP and consisted of lava flows erupted along NW–SE trending fractures; the second (Mursia Basalt) represented the island's most recent volcanic activity (younger than 10 ka BP).

Sampling was planned according to the volcanic units described in the literature and concerned all the basalts and the post-Green Tuff units. Only three of the pre-Green Tuff silicic units were sampled, because they are hard to reach. Most units were sampled at two sites, and samples were collected over a wide area to check possible post-emplacement movements of the large disrupted blocks, which are typical of pantellerite lava flows. A total of 150 samples,

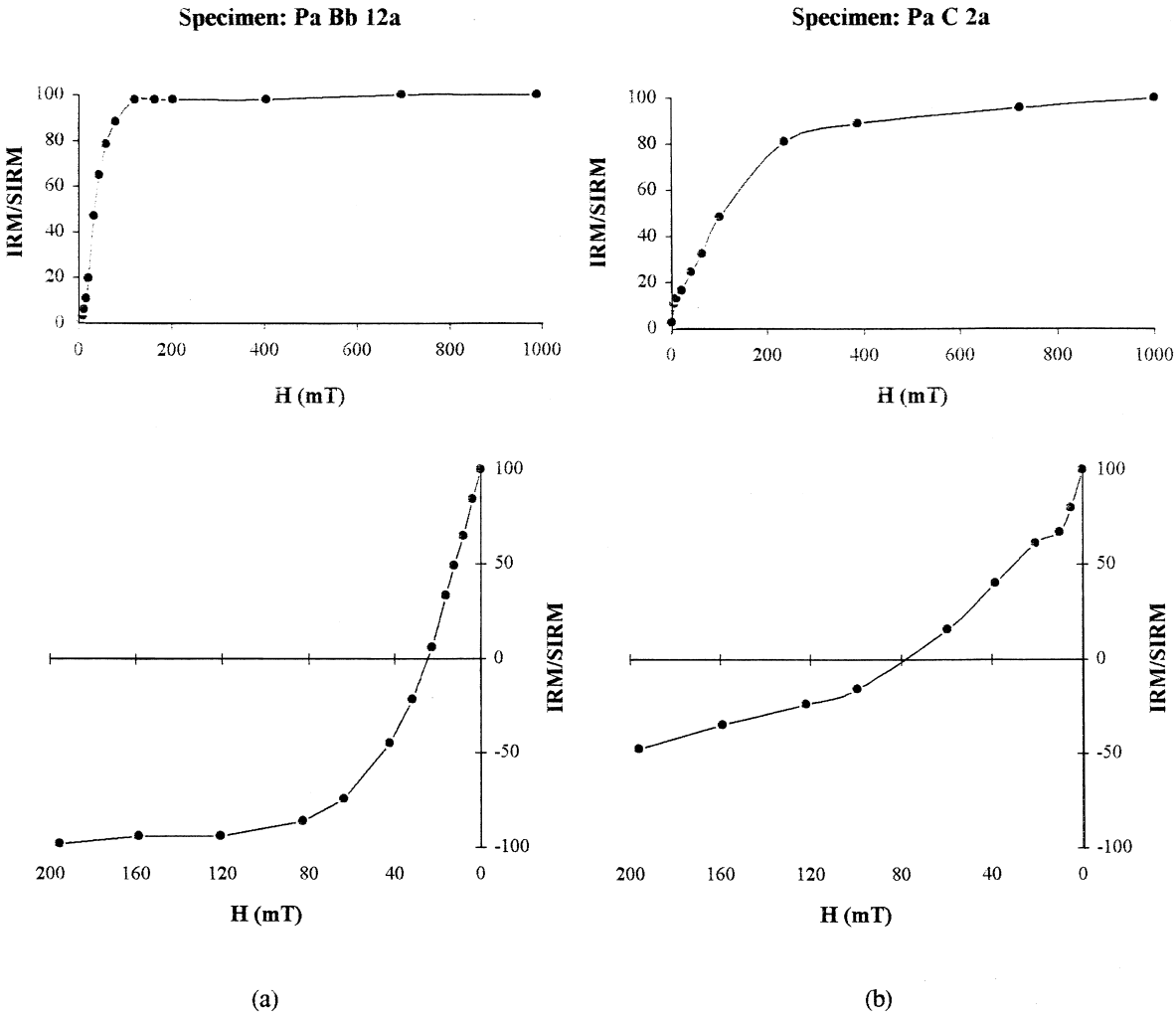


Fig. 3. Isothermal remanent magnetisation (IRM) acquisition curves and back-field removing curves. (a) Punta Alca basalt. (b) Green Tuff.

oriented by both magnetic and solar compass, were cored or collected as hand samples at 30 sites.

3. Magnetic measurements

Two or three specimens were cut from each core and hand sample and their magnetic susceptibility and anisotropy measured with a KLY-2 bridge. Natural remanent magnetisations (NRMs) were measured with a JR-4 spinner magnetometer.

Magnetic properties vary widely as a function of lithology (Fig. 2). The NRM intensity and the magnetic susceptibility, k_m, of the basalts are one to two orders of magnitude larger than those of peralkaline rhyolites. The k_m of the trachytes is similar to that of the basalts, but their remanent magnetisations, J_r, are lower. With the exception of two peralkaline rhyolite sites, J_r is always larger than the induced magnetisation, J_i, computed for a regional field value of $H = 35$ A/m (Meloni et al., 1988).

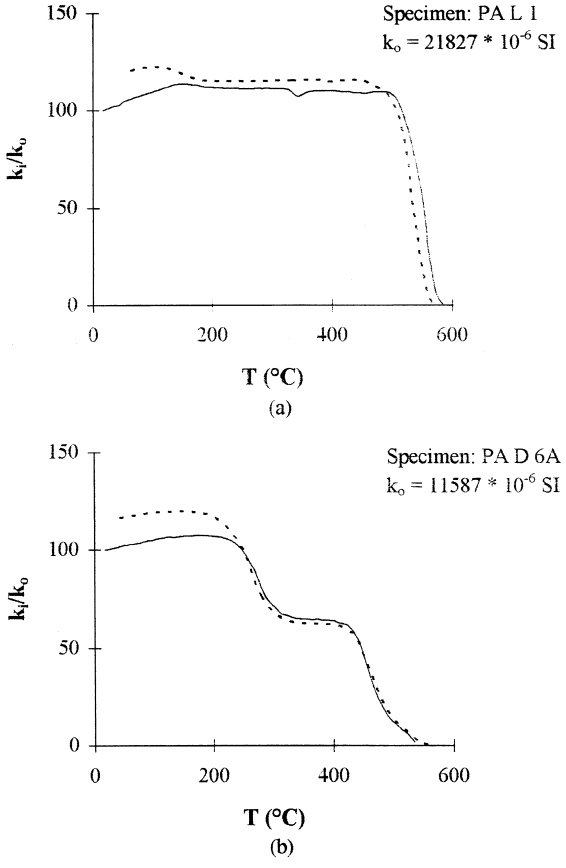


Fig. 4. Magnetic susceptibility in function of temperature curves. Symbol: full line = heating, dashed line = cooling. (a) Mursia basalt. (b) II Silicic cycle trachyte. k_0 = Room temperature susceptibility value.

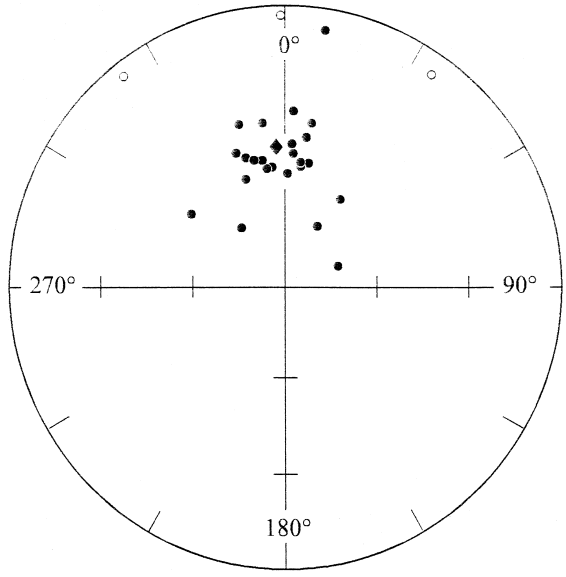


Fig. 5. Equal-area projection of NRM directions. Symbol: full/open dots = positive/negative inclinations, diamond = Fisher's mean direction.

Magnetic mineralogy was investigated by measuring the isothermal remanent magnetisation (IRM) and the susceptibility as a function of temperature. The IRM acquisition and back-field curves may be subdivided into two groups (Fig. 3). The first includes all the basaltic rocks: saturation is reached at an applied field value of 0.1 T and the remanent coercive force ranges from 20 to 40 mT. All other lithologies fall within the second group. The IRM acquisition curve displays an inflection at about 0.1 T; afterwards, the intensity increases again and saturation is only approached at the maximum field available (1.5 T). The back-field curve shows a kink in the trend and the remanence coercive force varies from 50 to 80 mT. The rocks of the two groups also show a different behaviour during the k vs. T experiments run using a Bartington MS2 furnace. The basalt curves are all similar and show a Curie point

between 570 and 600°C (Fig. 4), whereas the Curie points of trachytes and peralkaline rhyolites fall in two ranges, one at 200 to 400°C and the other at ≈ 550°C. Both points are found in some rocks. These results support the occurrence of high-Ti titanomagnetite to magnetite, and its weakly oxidised products, as main carriers of magnetisation. They agree well with the microprobe and X-ray fluores-

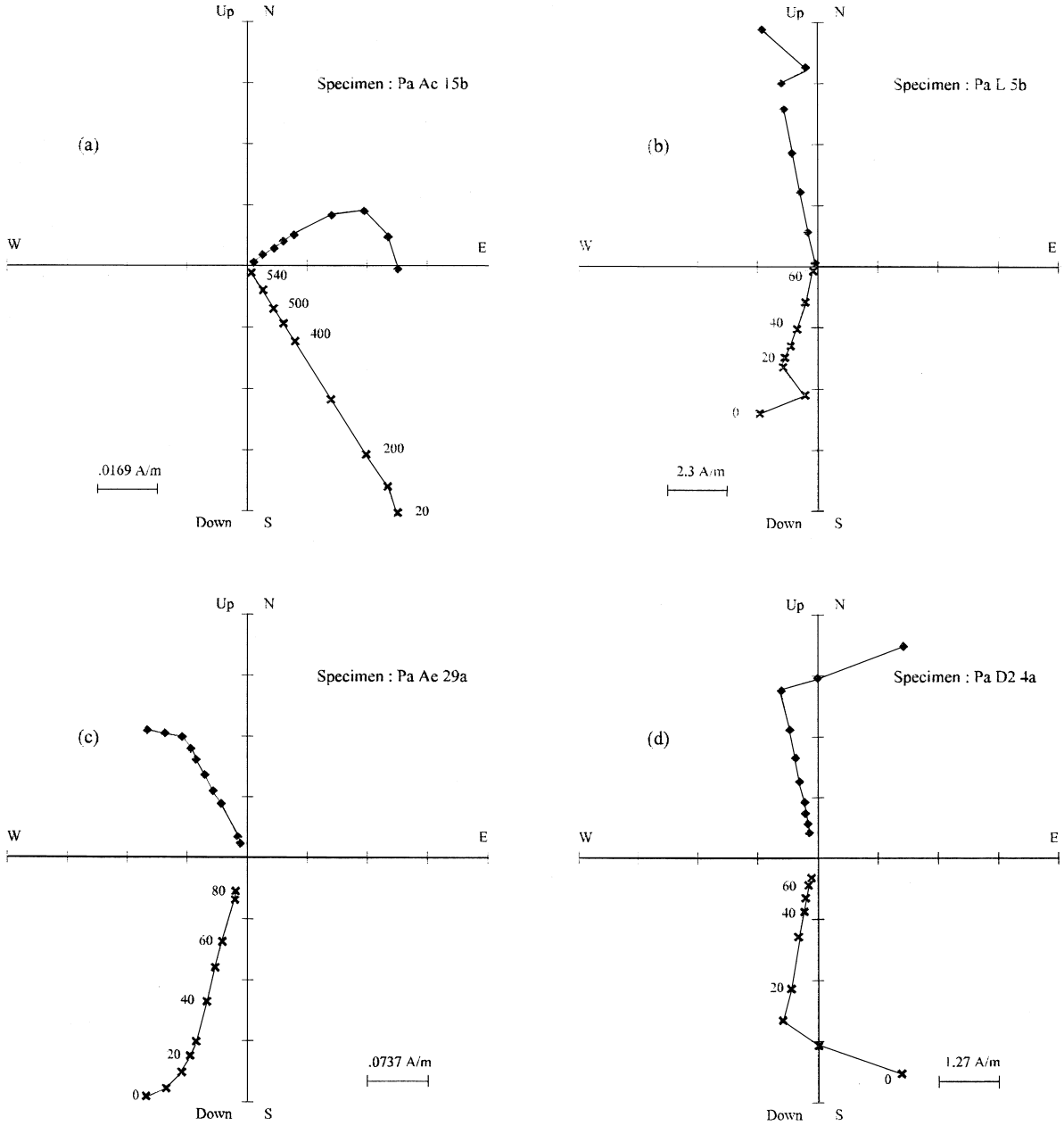


Fig. 6. Zijderveld diagrams of thermal (a) and Af demagnetisation (b–c–d). Symbol: diamond = declination, cross = apparent inclination.

cence data from the literature (Pignatelli, 1994), which show occurrence of the Fe-oxides of the magnetite-ulvöspinel series with a wide range in Ti content.

The grain size is mainly PSD and SD, because the median destructive field (MDF) ranges from 20 to 70 mT and the Koenigsbeger ratio values from 3 to 57. MD grains have been recognised in the trachyte of the II silicic cycle (MDF = 5 mT, ratio $H_{cr}/H_c > 4$ and $M_{rs}/M_s < 0.1$).

Three sites were excluded, two because of their unusual high intensity, probably due to lightening and the third because it was sampled in the tilted block of 'Montagna Grande' (Orsi et al., 1991a), a rotated direction confirmed the tilting. The NRM directions, as shown in the equal-area projection of Fig. 5, are rather scattered; the mean site magnetic inclinations in particular are weakly negative to almost vertical. On the whole, the mean NRM shows normal polarity directions, consistent with the emplacement ages of the volcanics during the Brunhes chron. At least one specimen per site was stepwise demagnetised in alternating fields (Af), up to 100 mT and one thermally, up to 600°C. Af demagnetisation was more efficient in removing secondary components and isolating the stable characteristic remanent magnetization (ChRM). After the first demagnetisation steps, its direction pointed straight towards the origin of the Zijderveld diagrams (Fig. 6), which show that demagnetisation at 20 mT peak field (30 to 50 mT for some peralkaline rhyolites) completely removes secondary components. This point was confirmed by systematic stepwise demagnetisation of three sites, one for each lithotype (basalt: L, trachyte: D2, peralkaline rhyolite: Ae). The mean site direction was calculated using the ChRM directions derived first after demagnetisation at 20 mT peak field, then from stable end-point analysis. The two mean directions were statistically indistinguishable and the null hypothesis of a common mean direction passed the McFadden and Lowes (1981) test at the 95% significance level.

4. Discussion

The mean ChRM declination and inclination computed for each unit are reported in Table 1, together with the corresponding virtual geomagnetic pole (VGP) position. They are also plotted as a function of age (Fig. 7) to show the trend of PSV. The reference axes in the figure ($D = 0°$, $I = 55°$) represent the values of the GAD at the latitude of Pantelleria. The time-scale has been reconstructed in the light of the radiometric data and stratigraphic constraints in the literature (Civetta et al., 1984, 1988; Mahood and Hildreth, 1986). All the post-Green Tuff units are associated with radiometric ages, whereas there are only a few geochronological data for pre-Green Tuff units. Both declination and inclination vary within the PSV range. Most inclination values, however, and all those post-dating the Green Tuff are lower than the GAD inclination. As a result, the mean paleomagnetic direction of Pantelleria, calculated from 27 site-mean values ($D = 358.5°$, $I = 46.9°$, $k = 30$, $\alpha_{95} = 5.2°$), differs significantly at the 95% confidence level from the GAD direction.

The PSV amplitude has been estimated by computing the angular dispersion of the VGPs about the geographic pole. The total angular dispersion is $S_T = 17.8°$, reduced to 17.1° after corrections for within-site dispersions, S_w (Doell, 1970). The total angular dispersion has been associated with a lower, $S_l = 15°$, and an upper value, $S_u = 21.9°$, calculated according to Cox (1969). The S_T variation is a function of the sampling site latitude, as shown in Fig. 8, which reports the VGP dispersions from Pantelleria together with data from the literature (Lund, 1985; Böhnel et al., 1990) and the theoretical curve calculated from the 'Model G' of the geomagnetic field by McFadden et al. (1988). The Pantelleria dispersion is about 1.5°–2° higher than the value expected from the model. The scatter for nearby volcanic zones in the Tyrrhenian area with a partially contemporary volcanic activity interval, such as the Aeolian Islands and Vulsini (Zanella, 1995), is, on the other hand, consistent with the model. The S_T computed with respect to the mean VGP of Pantelleria (PLat. = 82°N, PLong. = 200°E) is 15.9°, which is consistent with 'Model G'. This discounts the possibility of inadequate time interval sampling or a statistically insufficient number of sites, and indicates that the mean VGP differs slightly from the GAD in Pantelleria. Even the presence of local tectonic movements is not an acceptable explanation, since different lithotypes, not spatially associated but

Table 1
Paleomagnetic directions recorded in Pantelleria volcanites (latitude = 36.8°N, longitude = 12°E)

Volcanic unit	Lithology	Site	N/n	J_r	ChRM				VGP		Age	References
					D	I	k	α_{95}	PLat.	PLong.		
II Basic cycle	Punta Mursia basalt	L	1/12	13.7	352.6	30.0	43	7.0	68	212	< 10 ka BP	[1]
VI Silicic cycle	Peralkaline rhyolite	I	2/16	0.2	25.9	31.3	71	9.3	60	136	9.0 ± 3.7 ka	[2]
V Silicic cycle	Peralkaline rhyolite	H	2/18	0.1	2.0	35.8	56	4.7	73	186	12.7 ± 2.7 ka	[2]
IV Silicic cycle	Peralkaline rhyolite	G	2/17	0.5	355.1	48.7	57	5.1	82	223	16.2 ± 3.0 ka	[2]
III Silicic cycle	Peralkaline rhyolite	F	2/18	0.2	355.5	50.5	93	4.1	83	227	23.5 ± 2.0 ka	[2]
I Basic cycle	Punta S. Leonardo basalt	E	2/20	6.3	342.5	50.0	13	10.5	74	265	27 ÷ 30 ka	[1]
II Silicic cycle	Gadir peralkaline rhyolite	D1	1/10	0.6	324.8	25.6	20	13.8	51	255	28.7 ± 4.2 ka	[2]
	Mt. Gibele trachyte	D2	2/18	1.1	1.6	40.8	70	4.6	76	186	32.4 ± 4.5 ka	[2]
I Silicic cycle	Green Tuffs	C	3/33	0.5	7.3	52.9	99	2.7	83	129	49.6 ± 2.0 ka	[3]
Syn-Green Tuff silicic cycles	Trachyte	Af	2/27	0.1	354.3	49.1	130	2.4	82	229		
	Peralkaline rhyolite	Ae	1/10	0.2	5.4	38.8	83	5.3	74	173	47.0 ± 4.0 ka	[4]
Pre-Green Tuff basic cycles	Punta Alca basalt	Bb	1/12	6.2	350.2	49.0	124	3.9	79	245	83.0 ± 5.0 ka	[1]
	Cuddia del Cat basalt	Ba	2/14	5.6	353.4	51.1	145	3.3	83	241	118.5 ± 9.0 ka	[1]
Pre-GreenTuff silicic cycles	Peralkaline rhyolite	Ad	1/9	0.4	6.1	62.9	280	4.0	81	42		
	Peralkaline rhyolite	Ac	2/16	0.3	348.7	52.5	82	4.1	80	263	124.0 ± 6.0 ka	[4]
	Peralkaline rhyolite	Ab	1/8	0.3	355.9	54.6	87	6.0	86	256	159.0 ± 8.0 ka	[4]

N/n = number of sites/number of specimens.
J_r = remanent magnetisation (in A/m).
D, I = declination and inclination.
k = Fisher's precision parameter.
α_{95} = Semi-angle of confidence.
PLat., PLong. = latitude and longitude of the VGP.
[1] Civetta et al., 1984.
[2] Civetta et al., 1988.
[3] Cornette et al., 1983.
[4] Mahood and Hildreth, 1986.

of the same age, often show similar magnetic inclinations (see for example, lithotype L, Mursia basalt, and I, pantellerite of the 6th silicic cycle).

The declination and inclination anomalies ΔD and ΔI, which express the difference between the observed values at a particular site and those expected from the GAD field, have often been used to describe the deviation from the GAD. The ΔI at Pantelleria is $-8.1°$. Negative/positive ΔI have been described in paleomagnetic investigations in the northern/southern hemispheres (Wilson, 1970, 1972; Merrill and McElhinny, 1977; Schneider and Kent, 1988). In these studies, the anomaly, indicated by Wilson (1970) as a 'far-sided' effect, has been attributed to small secondary effects interpreted as a manifestation of long-term non-dipolar components that slightly deviate the time-averaged field from the GAD configuration. Johnson and Constable (1995, 1996) have confirmed the existence of these anomalies from the volcanic records of the last 5 Myr, and developed an interpretative model. The experimental data and the models (Merrill and McElhinny, 1977; Schneider and Kent, 1988; Johnson and Constable, 1995), show lower ΔIs than those recorded in Pantelleria at the same latitude. Furthermore, at Pantelleria, the very low inclinations are mainly recorded in the post-Green Tuff units, as can be seen from comparison of the mean ChRM directions, for post-

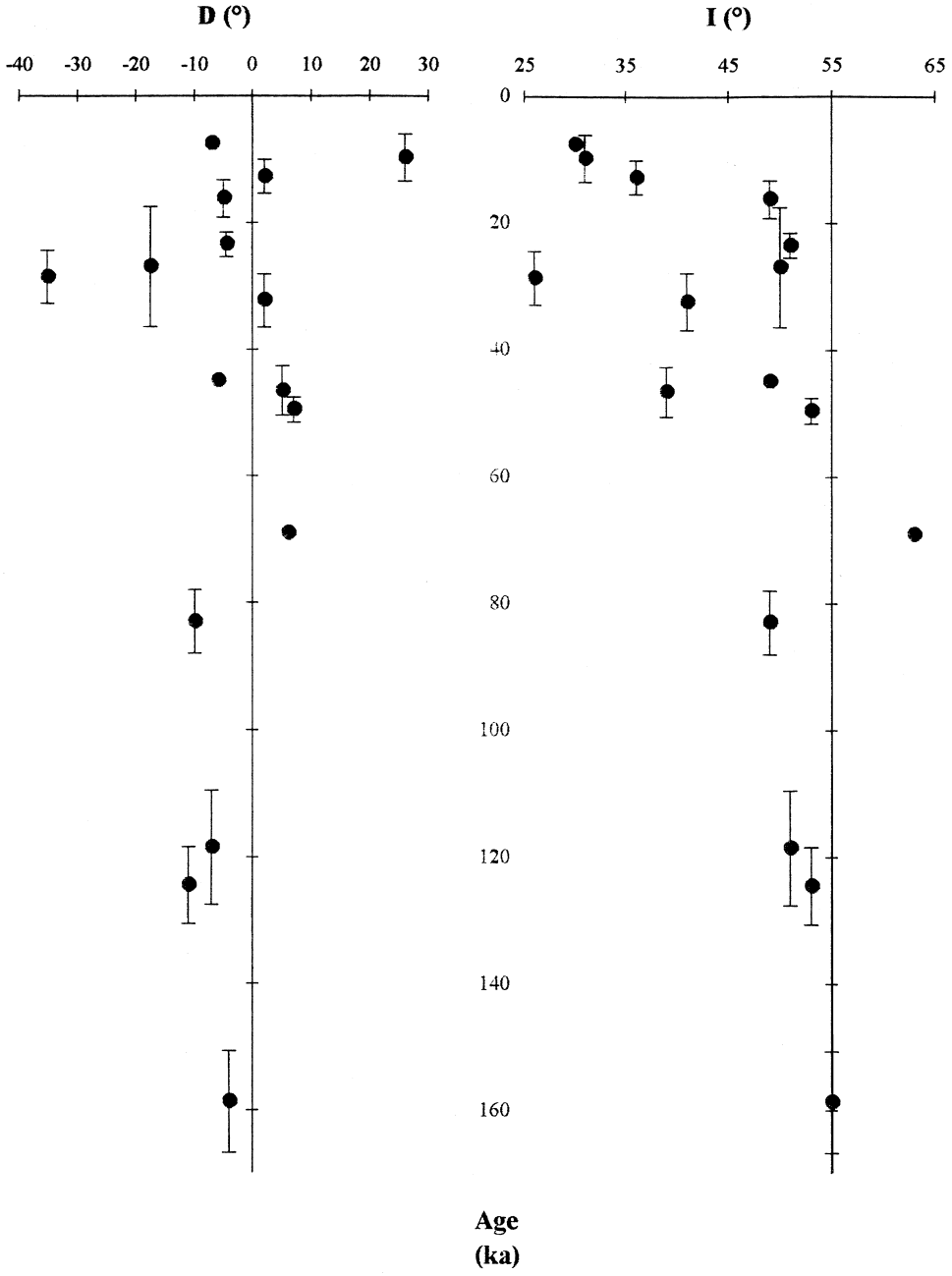


Fig. 7. Variations of paleomagnetic directions (ChRMs) in function of age of the Pantelleria volcanites. Symbol: dots = lithotype means. The error bars are equal to the age uncertainties.

and pre-Green Tuff units in Table 2, and in the equal-area projection of Fig. 9. Fisher statistics, in fact, show that the dipole and pre-Green Tuff mean inclinations are the same, whereas the post-Green Tuff units have a ΔI of $-10.6°$. This result does not significantly change ($\Delta I = -8.6°$), discarding the

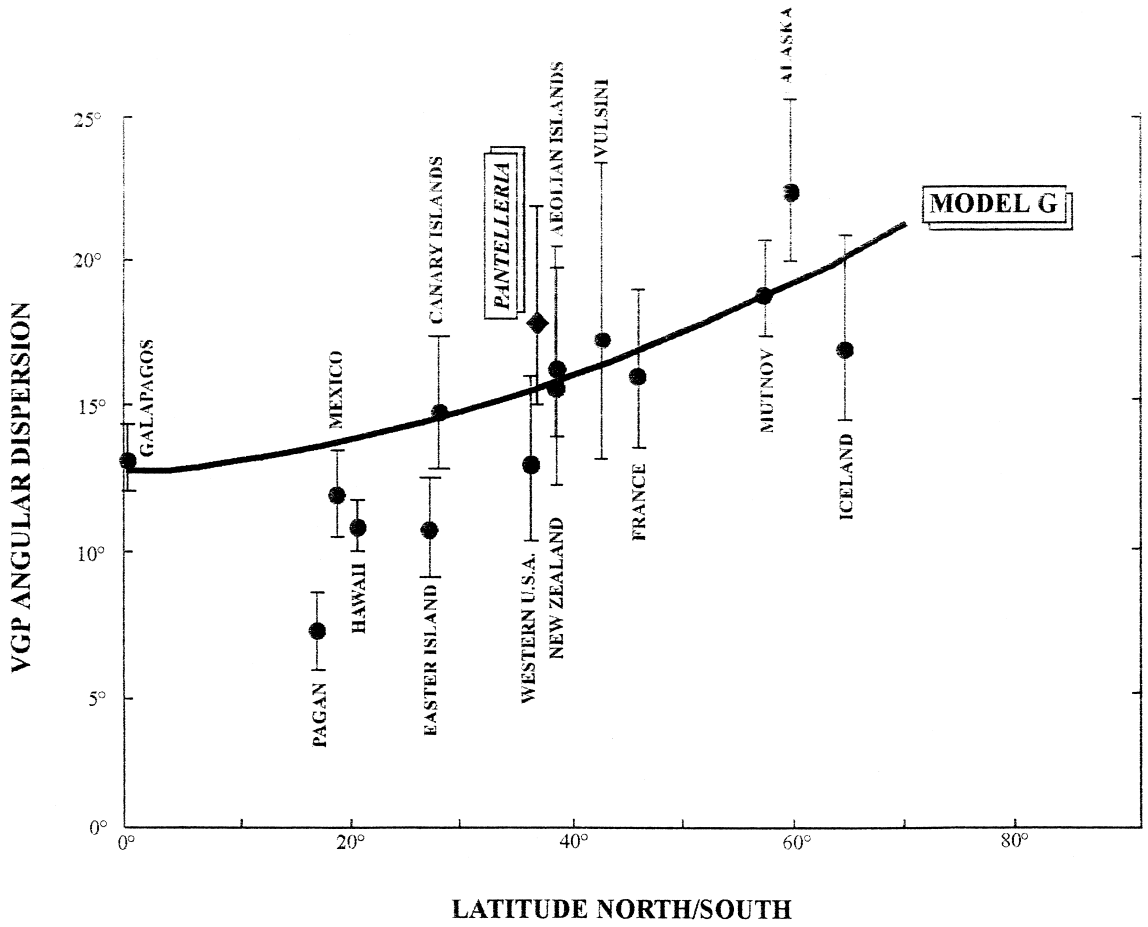


Fig. 8. Curve of the variation of angular dispersion of VGPs as a function of site latitude (model G of McFadden et al., 1988). Symbol: diamond = Pantelleria, circles = literature data (from Lund, 1985; Böhnel et al., 1990; Zanella, 1995).

data from the volcanic units, I and D1, which are characterised by the lowest VGPs latitude and whose significance might be challenged on the grounds of the large α_{95} values. In contrast, no systematic deviations have been found in the declination anomaly. The dipole and pre-Green Tuff paleomagnetic directions are fully comparable, although the data refer to only a few units. Those of syn- and post-Green Tuff units are always low. Some hypotheses to explain the inclination anomaly can thus be advanced.

(1) Excursion or 'over-active' PSV: the rocks might have been magnetised during either a geomagnetic excursion or a time span characterised by high-amplitude PSV fluctuations. Some excursions in the last 50 ka have been reported, such as Mono Lake (Denham and Cox, 1971; Liddicoat and Coe, 1979) and Laschamp (Bonhommet and Zähringer, 1969; Levi et al., 1990), and high-resolution paleomagnetic data recorded by lacustrine and marine

Table 2
Mean ChRM directions and inclination anomalies ΔI of all, post- and pre-Green Tuff units

Sites	n	D	I	k	α_{95}	ΔI
All sites	27	358.5	46.9	30	5.2	−8.1
Pre-Green Tuffs	7	353.2	53.5	162	4.8	−1.5
Post-Green Tuffs	20	359.9	44.4	25	6.7	−10.6

See Table 1 for the legend.

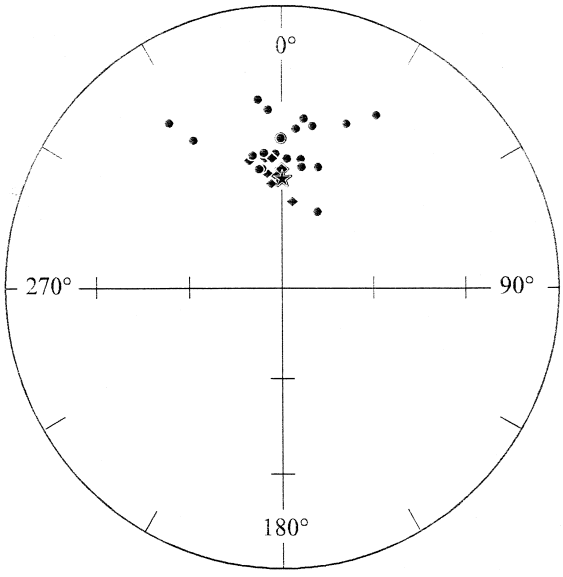


Fig. 9. Equal-area projection of ChRM directions. Symbol: dots = post-50 ka directions and Fisher's mean direction; diamonds = pre-50 ka directions and Fisher's mean direction; star = GAD at Pantelleria.

sediments have shown period of 'over-active' PSV. Lund et al. (1988) detected a quasi-periodic PSV waveform persisting for more than 15 kyr in the Wilson Creek lacustrine sediments above the Mono Lake excursion (ca. 28–27 ka BP), and attributed this behaviour to a kind of long-term memory in the geodynamo process. Levi and Karlin (1989) related to a reduced dipole moment high-amplitude inclination fluctuations recorded between 50 and 20 ka BP in a paleomagnetic record from a marine sediments in Gulf of California. At Pantelleria, the paleomagnetic directions are not so strongly deviated from the GAD (Table 1), as for a geomagnetic excursion, since the VGP latitude is always above 45°. The possibility of higher PSV fluctuations is difficult to test in the lack of a continuous record; moreover, no evidence of this has been found in nearby volcanic areas, such as Aeolian Islands (Zanella, 1995), where numerous directions for the last 50 kyr are available.

(2) Time undersampling: following Johnson and Constable (1995), a time interval of the order of 10^5 yr should be sufficient to average out any secondary effect of the geomagnetic field and obtain an estimate of the average field morphology, whereas sampling of several lava flows erupted over shorter time periods at a given site may not be sufficient. Biased field values may thus be attributed to inadequate sampling. Beck (1997) has shown that under-sampling can introduce a small bias towards shallow inclinations, especially at low latitudes. The probability of obtaining a shallow inclination is 0.55 if his Table 1 is applied to Pantelleria.

(3) Tectonic tilting: the low inclination values may be the result of tectonic tilting. They are observed all over the island and relate to different coeval lithotypes. Tilting of 10°–15° around an E–W axis would be required to explain the inclination anomaly, whereas the geological data show only 2° around a NE–SW axis. Furthermore, the tilting hypothesis cannot explain the difference between the pre- and post-Green Tuff data.

(4) Terrain effects: the deflection observed in Pantelleria cannot be the result of a magnetic terrain effect, because the induced and remanent magnetisation of the silicic rocks forming the bulk of the island are too low to produce a detectable deflection. Baag et al. (1995) have shown how the regional structures of a volcanic area in Hawaii can influence the geomagnetic record by producing a ΔI systematically shallower than the International Geomagnetic Reference Field (IGRF). They found that rocks with high susceptibility, namely basalts, induced variations in the measured magnetic field and questioned whether the PSV records could also have been influenced by regional structures. The typical susceptibility and intensity of the Hawaiian example ($k_m \approx 0.126$ SI and $J_r \approx 10$ A/m), however, are substantially higher than those for the bulk of Pantelleria.

(5) Regional anomaly of the geomagnetic field: the aeromagnetic map of Italy shows magnetic anomalies of 1500–2000 nT around Pantelleria. The possibility that a large magnetic body at depth may be responsible for this has been investigated by modelling a SW–NE profile across the island from the map of its residual field at 1000 m above sea level. The 0.8-km thick sequence of basaltic lava flows and hyaloclastites found by the geothermal exploration drill, PPT2, 200 m below sea level (Pignatelli, 1994) has been assumed as the source of the anomalies, because measurements on recovered samples have shown high susceptibility and NRM values ($55\,000 \times 10^{-6}$ SI and 2 A/m, respectively),

while those from the comendite intrusion found by drill PPT1 under Montagna Grande are too low (6000×10^{-6} SI and 0.5 A/m) to affect the anomalies. The model for the residual intensity at 1000 m derived from the 3-D algorithm of Coggon (1975) was used to calculate the vertical and horizontal component anomalies at 200 m above sea level, considered as the mean value for the topographic surface. The ΔI discrepancies calculated from these components were found to be always within the range $-1°$ to $1°$, i.e., well below the $-8.1°$ found for the island.

5. Conclusion

This paleomagnetic study of volcanics of Pantelleria has provided 16 time-independent directions that all vary within the PSV. The mean ChRM direction, computed from 27 sites for a time interval of 150 kyr ca., does not average out the PSV; the inclinations are low as compared to the GAD and the ΔI is $-8.1°$. Lower inclination values are mainly related to units younger than 50 kyr. Both tectonic movements and local geomagnetic anomalies fail to explain this ΔI anomaly and it may therefore be reasonably related to a regional long-lived, non-dipolar feature of the geomagnetic field lasting for the last 50 kyr. Long-term secular variations of the geomagnetic field of 40–50 kyr periodicity have been recognised by Yamazaki and Ioka (1994), and possibly ascribed to orbital forcing. Low inclination data have also been reported by Holt et al. (1996) in a study of lava flow cores from Hawaii for the past 400 kyr and attributed to a significant non-dipole field contribution.

Acknowledgements

I am especially grateful to R. Lanza for helpful discussion and constructive criticism, to G. Orsi for his help in planning the sampling and to A. Sbrana for making available the results of the geothermal exploration program. I wish to thank J.D.A. Piper and an anonymous reviewer for the improvements to the manuscript. This research was supported by University of Torino 60% fund.

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